# Electrical Workshop II

# Chapter Two

## Design and Winding of Small Power Transformers

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# *Contents*

- **Transformer definition purpose, properties and type;**
- **Necessary parameters need to be considered during transformer design;**
- **Design of sheet pack, winding coil and winding assembly of transformer;**
- **Winding of coils (primary & secondary) using winding machine;**
- **Testing of the rewind transformer using different testing mechanisms to check its efficiency**

# *Objectives*

**After studying this unit, the student will be able to**

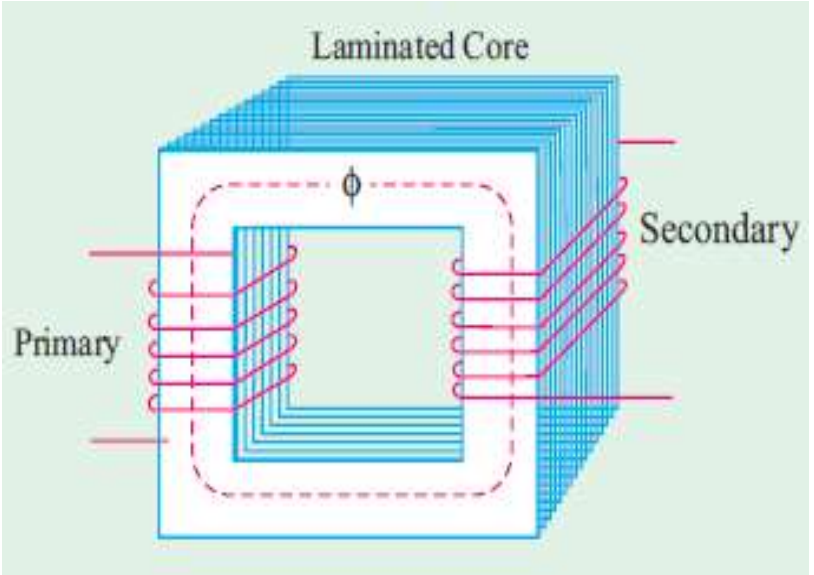
- **State the purpose of a transformer.**
- **Construct transformer practically in the workshop.**
- **Identify the common types of transformers from their schematic diagrams.**
- **Design small power transformer using design parameters.**
- **Choose the proper transformer taps to obtain the desired output voltage.**
- **Read transformer winding diagrams and connect a transformer for the desired primary and secondary voltage.**
- **Test re-winded transformer using different testing mechanisms**

# *Purpose of Transformer*

- A transformer is a static (or stationary) piece of apparatus by means of which electric power in one circuit is transformed into electric power of the same frequency in another circuit.
- It can raise or lower the voltage in a circuit but with a corresponding decrease or increase in current.
- ***The purpose of a transformer is to change electrical voltage to a different value.***

- The method of transferring electrical energy by a transformer is done indirectly.
- The physical basis of a transformer is **mutual induction** between two circuits linked by a common magnetic flux.
- In its simplest form, it consists of **two inductive coils** which are electrically separated but **magnetically linked through a path of low reluctance**.
- The two coils possess high mutual inductance.
- If **one coil** is connected to a **source** of **alternating voltage**, an **alternating flux** is set up in the **laminated core**, most of which is linked with the other coil in which it produces mutually-induced e.m.f. (according to Faraday's Laws of Electromagnetic Induction $e = M\, di/dt$).

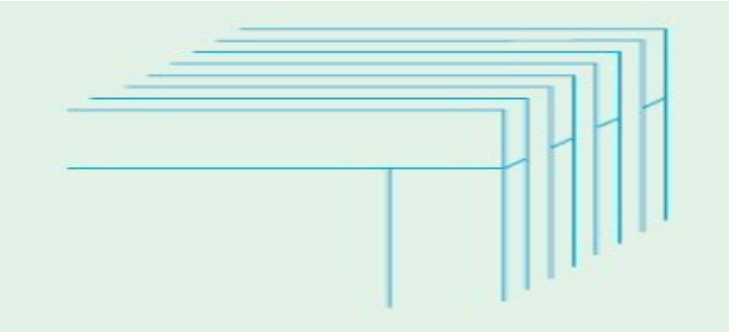
- If the **second coil circuit is closed**, a current flows in it and so electric energy is transferred (entirely magnetically) from the first coil to the second coil.
- The first coil, in which electric energy is fed from the a.c. supply mains, is called **primary winding** and the other from which energy is drawn out, is called **secondary winding**.



# *Transformer Construction*

- The **simple elements** of a transformer consist of **two coils** having mutual inductance and a **laminated steel core**.
- The two coils are **insulated** from each other and the steel core.
- Other necessary parts are : some suitable container for **assembled core** and windings ; a suitable medium for insulating the core and its windings from its container.
- The core is constructed of transformer **sheet steel** laminations assembled to provide a continuous magnetic path with a minimum of **air-gap** included.

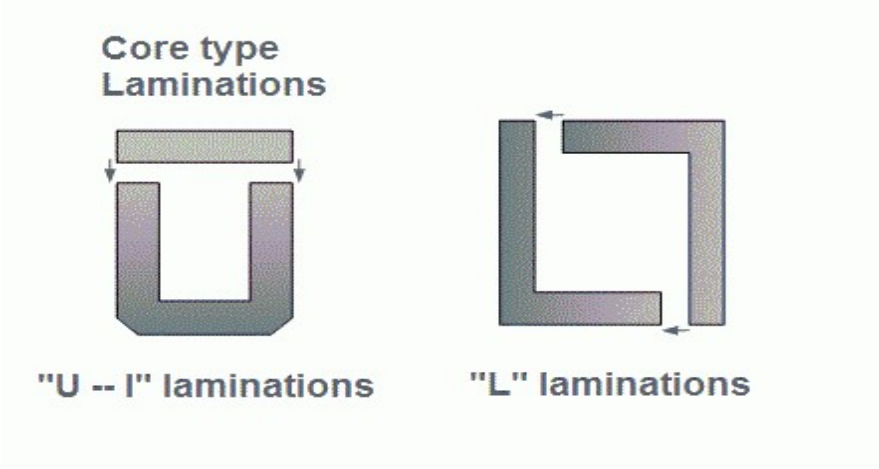
- The core laminations (in the form of strips) are joined.
- The joints in the alternate layers are staggered in order to avoid the presence of narrow gaps right through the cross-section of the core.
- Such staggered joints are said to be **'imbricated'.**

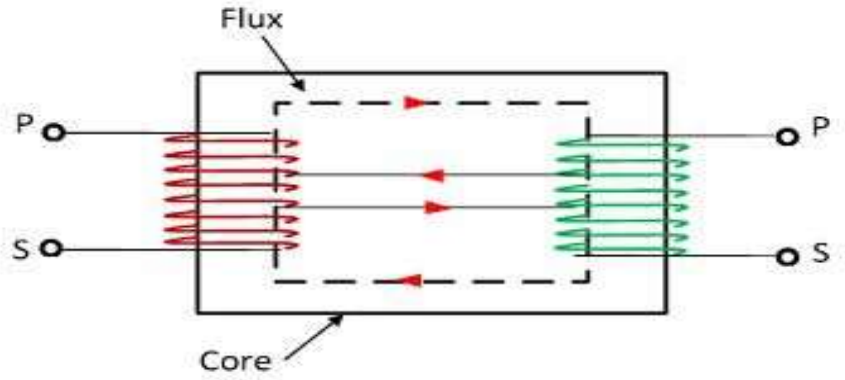


- Constructionally, the transformers are of **two** general types;

  **(*i*) core-type** and **(*ii*) shell type**

- **Core type transformers,** the ***windings surround*** a considerable part of the core
- In the simplified diagram for the core type transformers, the primary and secondary winding are shown located on the **opposite legs** (or limbs) of the core, but in **actual construction**, these are always ***interleaved*** to reduce leakage flux.
- Half the primary and half the secondary winding have been placed side by side or concentrically on each limb, not primary on one limb (or leg) and the secondary on the other.

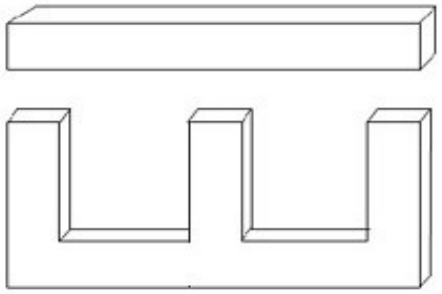


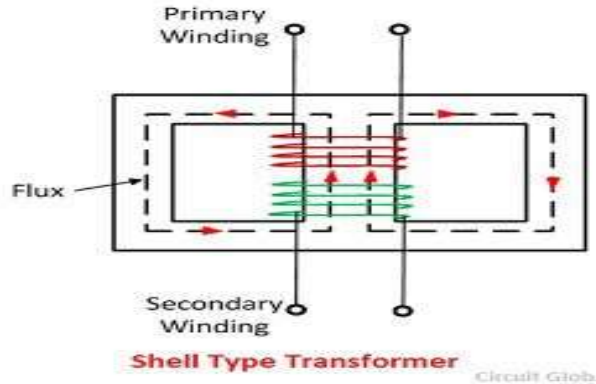




Core Type Transformer

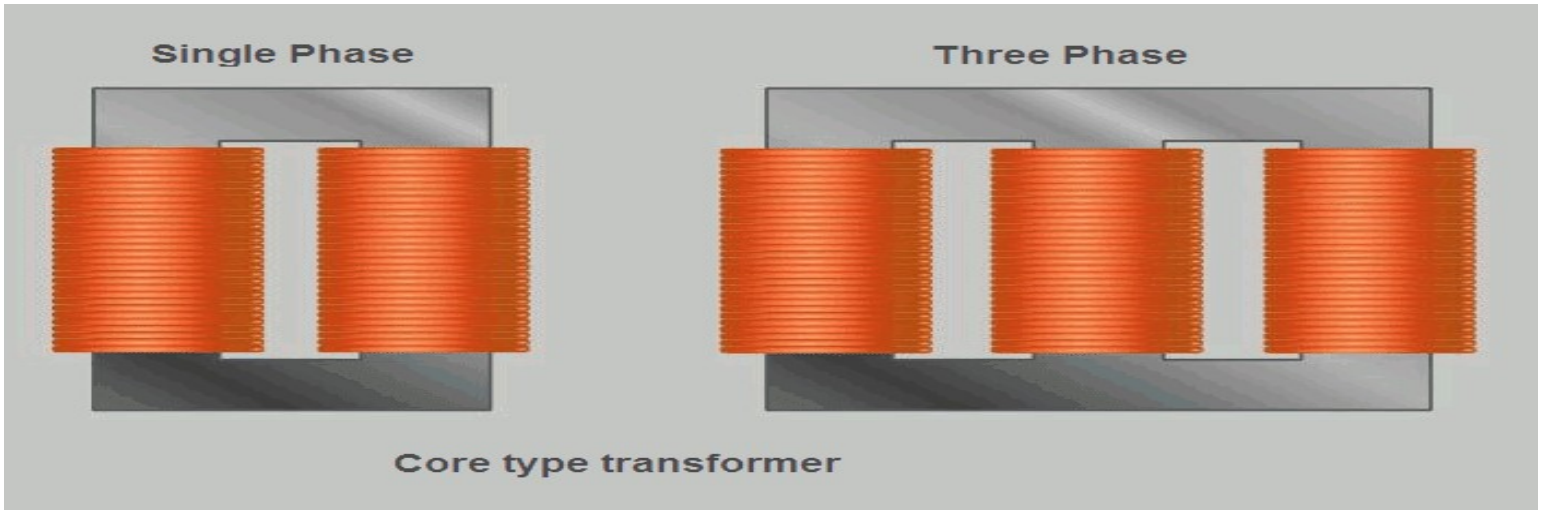
Circuit Globe

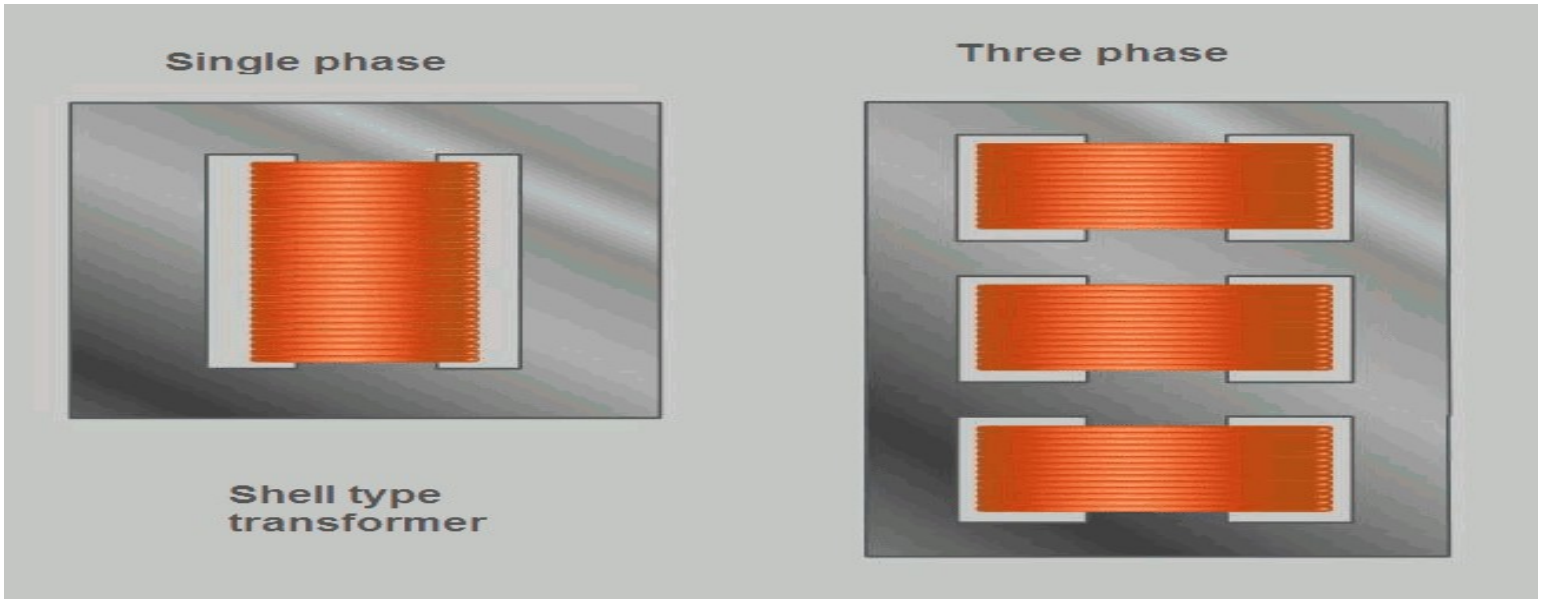
- **Shell type transformer**, the ***core encircles*** the winding of the transformer.
- The laminations are cut in the form of a long strip of **E**'s, and **I**'s as shown in the figure below.
- To reduce the high reluctance at the joints where the lamination are butted against each other, the alternate layers are stacked differently to eliminate continuous joint.
- The shell type transformer has three limbs or legs.
- The central limb carries the whole of the flux, and the side limb carries the half of the flux. Hence the width of the central limb is about to double to that of the outer limbs.





Shell Type Transformer


Single Phase
Three Phase
Core type transformer


Single phase
Three phase
Shell type
transformer

**Types of transformers :**

- **Step up/Unit transformers** – located at output of a **generator** to step up the voltage level to transmit the power.
- **Step down/Substation transformers** – Located at main distribution or secondary level transmission substations to lower the voltage levels for distribution 1st level purposes.
- **Distribution Transformers** – located at small distribution substation. It lowers the voltage levels for 2nd level distribution purposes.
- **Special Purpose Transformers (Instrument Transformers) -** E.g. Potential Transformer (**PT**) , Current Transformer (**CT**).

# Power Transformer Design

- The designer first needs several known factors to design a transformer.
- For a transformer using a **sine or square wave**, one needs to know the ***incoming line voltage, the operating frequency, the secondary voltage(s), the secondary current(s), the target efficiency, the physical size one can use, and the cost limitations(optional in the workshop).***
- Once these factors are known, design can begin.

## 1. Initial calculations

- The designer first starts with the **primary voltage** **and** **frequency**. Since they are a known factor, they are the first numbers to be plugged into the equations.
- One then will find the **power in watts (or volt-amperes)** of each secondary winding(for 3 phase) by multiplying the voltage by the current of each coil.
- These are added together to get the total power the transformer must provide to the load(s).

- The **transformer losses** in watts are estimated and added to this sum to give a total power the primary coil must supply.
- The losses are from wire resistance ($I^2R$ loss), loss in the core from magnetic hysteresis and from eddy currents. These losses are dissipated as heat.
- Next, one **selects the type of iron** by what efficiency is stated, and the value of losses to the user.
- Once the iron is selected, the **flux density** is selected for that material.

**2. Type of iron (electrical steel)**

- Each type of iron (steel) has a maximum flux density it can be run at without saturating.
- When using the equations, the two most important are the number of turns (**N**), and the core area (**A**) in **square centimeters or inches**, and match it to the total power in watts or volt-amperes.
- The **larger the core**, the **more power** it will handle.
- Once this **core size** is calculated, one then finds the **number of turns for the primary.**
- One then is looking at a transformer whose primary voltage will cause a flux density of a specified amount due to the number of turns in a certain type/size of core.

## 3. Secondary turns calculation

- Once the number of turns of the **primary** are calculated, the secondary windings numbers can be calculated with the **same turns per volt**.
- In reality, there are losses that have to be added, a rule of thumb is to allow for **5%** in losses (drops for voltage). (Transformers below 300W often have higher regulation losses).
- It is best to have a slightly higher voltage than one too low. Beware, smaller transformers which have a **higher turns per volt**, have higher losses, and the efficiency drops as the size goes down.

## 4. Thickness of coil windings

- When calculating the **coil thickness**, several things need to be considered.
- The voltage that each winding sees will determine the wires insulation thickness.
- Once this voltage is known, the **diameter** of the selected insulated wire can be used.
- By knowing the wire diameter, the number of turns per layer can be calculated, and the number of layers by using the ***window height and winding margins***.

- The **windows** are the openings on either side of the core.
- The window area is simply found by **multiplying** the **window width by its height**.
- Next, one adjusts the **thickness of the insulation paper** for the layers of each winding due to the voltage between the coils.
- This thickness is added to the total coil thickness by multiplying the paper thickness by the number of layers.

## 5. Insulation

- The insulation materials used in transformer construction varies, but are mainly the finish applied to the laminations, varnish or coated insulation on the wire, paper, or plastic film for the coil, and the varnish that the completed transformer is dipped into at the end of construction.
- The varnish dip that is applied, is generally done in a vacuum chamber. Most **varnish is a clear color, but some is dyed black**.

## 6. Wire selection

- The wire generally used in small transformer coils is **magnet wire**.
- **Magnet wire** is generally a solid copper wire with an insulating jacket such as varnish.
- Aluminum is sometimes used in power transformer windings to reduce costs.
- The connections on smaller transformers are ***soldered***, but larger transformer connections, especially ones handling high current, are ***brazed or welded***.
- The wire is selected by its ***ability to carry the proper amount of current*** without getting hot enough to melt the insulation or wire itself.
- It is sized by its ***cross sectional area*** measured in **circular mils per ampere**, or more commonly in engineering design circles, **amperes per square meter**.
- In transformer use, the circular mils per ampere runs anywhere from 500 cir. mils for intermittent operation, to 1500 cir. mils for heavy duty continuous operation. For most applications, 800 to 1000 cir. mils is a good starting point.

## 7. The core stack

- The **core stack** is the total amount of the steel laminations needed to produce the correct core area for the power in watts, or volt amperes that the transformer is required to handle.
- The core can be stacked in **two different ways** when using steel laminations.
  - The most common is the **interleaved fashion** where each lamination is staggered opposite to the other (**turned 180 degrees opposite the other**). This provides for the least amount of air gap in the core, and the highest efficiency.
  - The other way is **butt stacked**. In this way, all the **E** type lams are stacked on one side, and all the **I** type lams are stacked on the other. This way though creates an air gap where the butt joint is created thus ***increasing the losses.***

## 8. Stacking Factor

- When using the formulas to calculate the core area, a stacking factor should be included.
- It varies by the thickness of the material, type, and whether it is either butt stacked, or interleaved. This factor ranges from around ***0.80 to 0.98***.
- The **stacking factor is needed** due to each iron lamination having a thin coating of insulation on each face. This coating thereby insulates each lamination from the other minimizing eddy currents.
- The stacking factor adjusts the core size due to the extra thickness of this insulative coating so that the proper amount of iron is used in the core.
- Without this factor placed into the equations, the core would be slightly under-sized in iron content, even though the core stack equaled the calculated measurement.
- The core in mention, not using the stacking factor, would then have a slightly higher flux density than what was wanted.
- ***It is the ratio of sum of pure steel length to the designed length of laminated core.***

## 9. Scrapless Laminations

- When **EI** and other laminations are produced, they are literally punched out of a coiled strip of electrical steel with a punch press.
- To reduce the amount of scrap as much as possible, the lamination is dimensioned in such a way that if **one placed two E** laminations with their ***open ends toward each other***, the two sections that make the **windows** would make the **two I's** needed when all four pieces are punched out at once.
  - Due to this, ***the window width***, and ***the leg width*** is generally **1/2 the width of the tongue**.
  - The ***window height*** would be equal to ***1/2 the width of the whole E lamination***.
- ***The lamination height would be equal to the window height plus the width of the window or leg.***

## 10. Core Stack Assembly

- There are several ways that the core stack is held together after assembly.
- One way, is using **mechanical fasteners**, but this can promote a problem, if the fastener is made of a magnetic material, and the screw hole in the lamination is too far into the flux path of the lamination.
  - If the magnetic flux cuts through a magnetic fastener, it causes the fastener to **heat up,** acting like a short-circuit, which in reality, is what happens, allowing the flux lines to ***mingle*** between the laminations as if they were shorted together.
  - To cure this, the fastener holes in the laminations are generally placed in the **far corner** of the laminations, where the least amount of magnetic flux travels. Also, magnetic fasteners have been used by ***adding a paper sleeve over the screw***, to insulate it somewhat, or a non-magnetic fastener, such as made from Aluminum or Brass, have been used.
- Some transformers, today, have a **welded seam** placed on the outside of the laminations to hold them together, but this can cause a ***small amount of loss***, the same way a magnetic screw would allow the magnetic flux to mingle between the insulated laminations.
  - Since it is on the outside surface of the lamination, though, and away from the majority of the magnetic flux toward the center and coil, the loss is generally low.

## 11. Watts versus Volt-Amperes

- A transformer's power handling ability is determined in **two different ways**.
  - If the transformer's **secondary** is supplying a totally **resistive load**, one can simply use **wattage**, or the voltage multiplied by the current.
  - However, if the transformer's secondary is supplying a **reactive load**, containing capacitance and or inductance, one must use **volt-amperes** in place of watts.
- **Volt-amperes**, (S) or (**VA**), is simply the RMS voltage times the RMS current, or, equivalently, wattage (P) divided by a power factor, or **P / PF = VA.**
  - ***For example:*** 1000 watts (P) / 0.80 (PF) = 1,250 Volt-amperes (VA).

# Design Parameters of Transformer

*(for detail design referee Lab Manual )*

- ***For designing a transformer, we need;***

1. ***Power rating***
2. ***Voltage levels (primary and secondary)***
3. ***Currents on both sides***
4. ***Primary and secondary coils wire diameter/size***
5. ***Iron Core area***
6. ***Numbers of turns (primary and secondary)***

- **As we are going to design small transformer (of small power rating) so we are neglecting core and copper losses as they doesn't matters in small transformers.**

# Calculations:

## 1. E.M.F. Equation of a Transformer

- Let $N_1$ = No. of turns in primary
  $N_2$ = No. of turns in secondary
  Φm = Maximum flux in core in Weber
  = $B_m \times A$
  $f$ = Frequency of a.c. input in Hz
- r.m.s. value of e.m.f/turn = 4.44 $f$ Φ$m$ volt/turn
- Now, r.m.s. value of the induced e.m.f. in the **whole of primary winding**
  = (induced e.m.f/turn) × No. of primary turns

$$E1 = 4.44\, f\, N_1\, \Phi m = 4.44\, f\, N_1\, B_m A$$

- *Similarly, r.m.s. value of the e.m.f. induced in secondary is,*

$$E2 = 4.44\, f\, N_2\, \Phi m = 4.44\, f\, N_2\, BmA$$

- *It is seen from the above two equations, that* $E1/N_1 = E2/N_2 = 4.44\, f\, \Phi m$. *It means that e.m.f./turn is the same in both the primary and secondary windings.*

# 2. Core Area Calculations:

Calculate area of core (central limb) by using following formula:

$$A = \frac{E1}{4.44 f B_m N} = \frac{1}{4.44 f B_m T}$$

***where,*** ***A- core area in sq cm (to be calculated)***

***f – operating frequency 50 Hz (known)***

$B_m$ ***– magnetic flux density in tesla (known)***

***N – number of turns.***

***T – turns per volt***

**Assumptions:**

- So, we know the ***frequency*** of the power system.
- We need ***magnetic flux*** and ***turns per volts***.
- For designing a small transformer magnetic flux density is averagely taken as 1 to 1.2
- By putting values we will get the area of core.

# 3. Selection of winding wires and strips

- Winding wires and strips are selected on the basis of the requirement of the **current** for which they have been designed.
- While selecting the size of a conductor, a designer must keep in mind the winding material as well as the current density.
- To start with the preliminary design, the current density (**J**) may be based on the following values:
  - For aluminum winding: 1.5 A/sq mm (**maximum**)
  - For copper winding: 3.0 A/sq mm (**maximum**)
- The rated current may be calculated on the basis of its **KVA**, number of phase and the rated voltage.
  - ❖ KVA = $\sqrt{3}$ * V* I (for three phase transformer)

$$I = \frac{KVA}{\sqrt{3} * V}$$

- Once the current is known, the conductor area may be calculated on the basis of assumed current density:

$$\textbf{Conductor area} = \frac{\textbf{current per phase}}{\textbf{current density}} = \textbf{I/J}$$

# Example

- **Assumptions:**
- For designing the small transformer take the frequency of 50hz, magnetic flux density of 1.2T, power rating of 50VA, turns per volt of 2.6, and current density of the copper is approximately 2.3A per mm square.
- **Step 1**: calculate the area of the core (central limb)

$$A = \frac{E1}{4.44 f B_m N} = \frac{1}{4.44 f B_m T} = 0.00144\text{m square}$$

- **Step 2:** Primary winding calculation

  - take the primary voltage as 230V, the primary current is:

    I1 = P/V1 = 50/230 = 0.218A

Let transformer we are going to design is 95 % efficient so,

I1 = VA/ (efficiency x V1) = 0.23A

**Primary Current** = 0. 23 Amp (approx.)

**Step 3: Number of Turns on primary**

- Total number of turns = turns per volt **x** primary side voltage = 2.6 x 230 = **600 turns**

**Step 4: Conductor Size**

- As, for copper the current density is taken as 2.3A per mm square So, for area of copper conductor

$$\textbf{Conductor area} = \frac{\textit{current per phase}}{\textit{current density}} = \textbf{I1/J}$$

$$= \textbf{0.23A/2.3A per mm square} = \textbf{0.1mm square}$$

**Step 5: Secondary winding Calculation**

Let assume that secondary voltage is 12V, then secondary current is:

$$\textbf{I2} = \textbf{50VA/12V} = \textbf{4.2A}$$

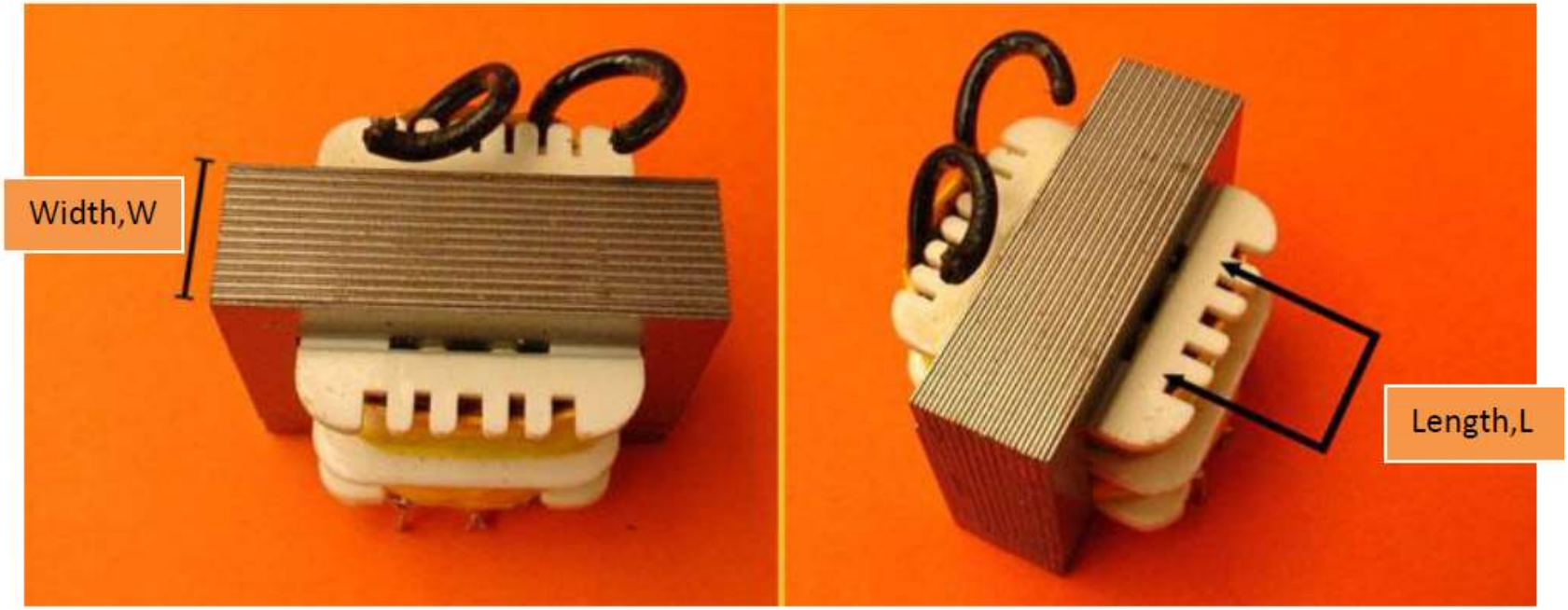
- So, area of secondary conductor is:

$$\textbf{I2/J} = \textbf{4.2A/2.3A per mm square} = \textbf{1.83mm square}$$

- Total number of turns = turns per volt **x** secondary side voltage = 2.6 x 12 = **32 turns volts**

## Step 6: Find the section kernel

- ***Section kernel*** $= \sqrt{P} = \sqrt{50} = 7.07$ cm$^2$
- *Thus, the size of the center core is 7.07* cm$^2$. *We can use any combination of length and width to get the area. For example, 3.535cm x 3.535cm = 7.07* cm$^2$.
  - **L= 3.535 cm, W= 3.535 cm**



***Thank you***

***for***

***your Attention!!***

